Goodwill After-Hours Donation System

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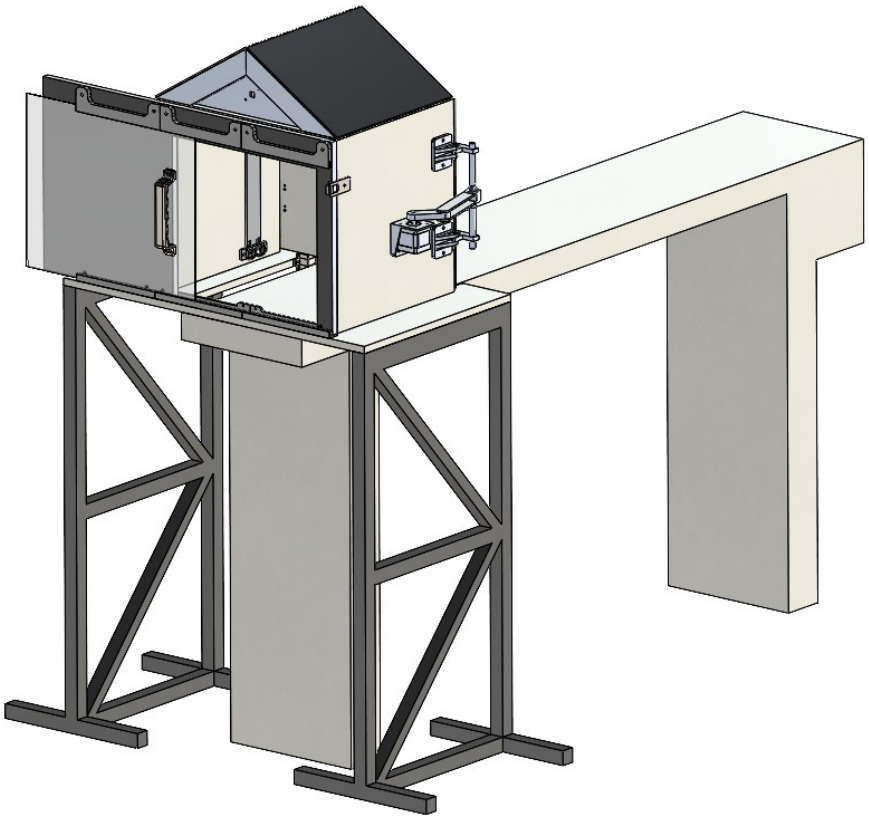


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1. Executive Summary:

Goodwill relies on community donations to support job training, sustainability, and workforce development programs. However, donations made after business hours are often left outside store locations, where they are exposed to theft, weather, and breakage. Fragile items such as glassware and electronics are especially vulnerable, reducing resale value and increasing waste and labor costs for staff. This project addresses that challenge by designing a secure, accessible after-hours donation system that protects donated items while remaining simple and intuitive for donors to use.

The final design is centered on a two-door interlock architecture that physically separates donor interaction from the secured collection space. Donors place items through an exterior, donor-facing door that provides an accessible entry point without allowing direct access to the interior enclosure. Once the exterior door is fully closed, the system verifies safety conditions and actuates an interior security door, transferring the item into a locked collection space. Item transfer is achieved using a level conveyor belt that moves donations horizontally rather than relying on gravity, eliminating vertical drops and reducing the risk of damage to fragile items.

Security is reinforced through a three-point locking strategy integrated into the interior door assembly. Two fixed locking points are located at the bottom left and bottom right corners of the interior door, where prying and outward bending forces are most likely to occur. These lower locks provide lateral constraint and distribute loads across the base of the door. A centrally located rotating lock is positioned at the top center of the interior door, directly between the exterior and interior panels. This top lock engages only when the exterior donor-facing door is fully closed, mechanically enforcing the two-door interlock and preventing simultaneous access to both compartments. Together, these three locking points limit door deformation under applied loads and reinforce the physical separation between donor interaction and the secured collection space.

Functional validation demonstrated that the system can reliably complete repeated donation cycles without jamming, unsafe access, or damage to donated items at the prototype scale. While the system has not undergone formal standards certification, it demonstrates a strong proof of concept. With production-grade materials, improved environmental sealing, and standards-based testing, the design could be realistically implemented at Goodwill locations as a secure, accessible, and durable after-hours donation solution.

2. Introduction:

2.1 Motivation

Across the United States, Goodwill plays a vital role in supporting community development through its resale programs, which fund job training and sustainability initiatives [1]. However, donation management outside of staffed hours remains a persistent challenge. At many Goodwill locations, donors leave items after hours near storefronts or donation doors, leaving them exposed to weather, theft, and accidental damage. According to Goodwill Michigan's 2024 Community Impact Report, the organization receives over 200,000 donations annually, with roughly 10% occurring after business hours [1]. Many of these after-hours donations are found broken or stolen leading to significant material loss and additional labor costs for staff.

A secure, reliable after-hours donation system strengthens Goodwill's mission by protecting donated goods, reducing waste, and ensuring equitable accessibility for all users.

2.2 Problem Statement

Current after-hours donation practices at Goodwill locations leave donated items vulnerable to theft, weather exposure, and accidental damage. Existing donation solutions either prioritize security at the expense of fragile item protection or rely on gravity-based chutes that damage delicate goods. As a result, many after-hours donations arrive broken, unusable, or stolen, increasing waste and labor costs while reducing resale value.

The problem is to design a secure, accessible, and durable after-hours donation system that allows donors to deposit items safely without staff present, prevents unauthorized access or retrieval, and protects fragile donations from impact damage during transfer.

3. Design Solution:

This section presents the final design solution for the Goodwill after-hours donation system. The design was developed through a user-centered and iterative engineering process that translated stakeholder needs into engineering requirements, evaluated existing solutions, generated and tested multiple concepts, and refined the system through prototyping and analysis. Each subsection builds toward an evidence-based justification of the final design and demonstrates how the proposed system addresses the limitations of current after-hours donation methods.

3.1 User Needs

Primary users include after-hours donors, who need a simple, low-effort, and safe deposit; Goodwill staff, who need intact, sortable items and tamper-resistant storage; and store neighbors and pedestrians, who benefit from a clean, quiet, and unobtrusive system. Secondary stakeholders include Goodwill leadership (loss prevention, brand trust, sustainability targets) and maintenance and installation contractors who require durable, weather-resistant components that are easy to service. These stakeholders were prioritized because their needs directly influence system safety, long-term usability, and the feasibility of deployment across multiple Goodwill locations.

User needs were informed by client discussions with Goodwill representatives, observation of existing donation practices, and review of prior donation system designs. These sources helped identify recurring challenges related to theft, damage to fragile items, and accessibility barriers during after-hours use. Stakeholders were prioritized based on their direct interaction with the system and their influence on operational success, including donors, Goodwill staff, and nearby community members.

Table 1: User Needs

Rank of importance	Abbrev.	User Need	Description
1	SEC	Security Against Theft/ Break-ins	Donors and staff need donations to remain secure from theft and tampering until collection
2	FRG	Protection of Fragile Items	Staff need fragile goods (glass, ceramics, electronics, etc.) to remain intact after deposit.
3	ACC	Donor Accessibility	Donors need the system to be easy to use, allowing them to deposit items without excessive effort
4	DUR	Durability	The system withstands repeated use, and exposure to outdoor weather conditions
5	CAP	Capacity	The system should be able to handle several donations through after-hours use.
6	CLN	Ease of Maintenance	The system should be straightforward for staff to clean and maintain.
7	NOI	Low Noise Operation	Operation should be low-noise to avoid disturbing nearby members of the community, especially during late-nights.
8	UNB	Unobtrusive Design	The system is intuitive and easy to use, or parking-lot flow, ensuring pedestrians and staff can move around it.
9	INT	Intuitive User Experience	The deposit process should be intuitive and require minimal instructions.
10	RPC	Standardizable	The design should be adaptable to various Goodwill locations nationwide.
11	CAP+	Continuous Donation Flow	The system should accommodate repeated after-hours drop-offs, ensuring donations don’t back up, jam or overflow.

Table 1 lists the full set of identified user needs. The most critical user needs are discussed below, as these directly drive key design decisions. Security is ranked highest because preventing theft protects donations and maintains donor trust, which is essential to Goodwill’s resale-based social mission [1]. Protection of fragile items follows, as glassware and electronics that arrive damaged cannot be resold, increasing waste and labor costs. Donor accessibility ranks third, emphasizing the need for an inclusive design aligned with ADA standards for equitable access [3]. Durability comes next since the system must withstand repeated use and outdoor exposure. Capacity ranks fifth, as it affects operational efficiency but is less critical to immediate safety and usability and is constrained in the current prototype.

3.2 Engineering Requirements and General Standards

To translate user needs into actionable design criteria, a set of engineering requirements was established and is summarized in Table 2. These requirements define measurable performance targets that guide design decisions and enable objective evaluation during testing.

Table 2: Engineering Requirements

Priority	Requirement (maps to)	Description	Units	Target	Rationale / Standard
High	Continuous item support (FRG)	Vertical drop distance during transfer	in	0	Eliminates free fall to protect fragile items
High	Operating reach (ACC)	Highest operable part (handle/slot centerline)	in	≤ 48	Accessible interface per ADA reach ranges [3]
High	Operating force (ACC)	Force to actuate door/slot	lbf	≤ 5	Low-effort actuation aligned with ADA guidance [3]
High	Anti-reach / anti-fish (SEC)	Second locked door prevents retrieval of deposited items	—	Pass	Physical anti-fishing features; no probe can contact contents
High	Tamper resistance (SEC)	Minimum wall thickness + lock	in	≥ 0.5 in steel + self-locking mechanism	Prevents tampering
High	Prying resistance (SEC)	Exterior Door and panel overall geometry	__	Pass	No permanent deformation or access under manual pry force
Med	Ingress protection (DUR/SEC)	Weather/particle ingress rating for enclosure	IP	IP54 (min)	Outdoor dust/splash resistance for electronics/interlocks [5]
Med	Capacity (CAP)	Internal volume for deposited items	gal	20 (5*)	Operational capacity target; prototype scaled down
Med	Noise level (NOI)	Sounds level during operation at 1m	dBA	<55	Limits disturbance to nearby pedestrians and residents
Low	Continuous donation flow (CAP+)	Time or cycles before system requires emptying	hr	≥ 24	Handles multiple drop-offs without overflow or jamming
Low	Material durability (DUR)	Resistance to corrosion and weather degradation	hrs	≥ 500	Follows ASTM B117-18 for outdoor exposure [2]

**Prototype scaled to 5 gallons due to build constraints.*

Donor accessibility is ensured by limiting the highest operable part to 48 inches and keeping the actuation force at or below 5 lbf, in accordance with the ADA Standards for Accessible Design [3]. Protection of fragile items is achieved by eliminating free fall entirely. Donations are transferred along a

level conveyor belt that maintains continuous support, preventing impact forces associated with vertical drops. Security is achieved through a two-door interlock system with anti-fishing geometry and tamper-resistant panels that prevent unauthorized access or retrieval of deposited items.

Durability and environmental reliability are addressed through an ingress protection requirement of at least IP54, ensuring resistance to dust and splashing water in semi-outdoor installations as defined by IEC 60529 [5]. Material durability expectations are informed by ASTM B117-18 guidelines for outdoor exposure [2]. Capacity targets were set at 20 gallons for a full-scale system, with the prototype scaled to 5 gallons due to fabrication constraints.

Low-noise operation was included as an engineering requirement to minimize disturbance to nearby pedestrians and neighboring properties during after-hours use. Continuous donation flow was addressed by requiring the system to operate for at least 2 hours without jamming or requiring emptying, supporting staff efficiency while avoiding excessive stacking that could increase damage risk.

Engineering standards play a critical role in ensuring the system is safe, accessible, and deployable across Goodwill locations. ADA standards govern donor interaction height and operating force, IEC 60529 applies to enclosure and electronics protection, and ASTM standards inform material selection and durability expectations. Compliance with these standards was considered during design and is evaluated through prototype testing and inspection [2], [3], [5]. To ensure compliance, prototype testing will verify that all measurable requirements meet defined thresholds, including actuation force, accessibility, and ingress protection performance in dusty environments.

3.3 Competitive Benchmarking

By comparing current market solutions against the most critical user needs, including security, protection of fragile items, accessibility, and durability, we can identify performance gaps and design opportunities. This benchmarking process focuses on how existing systems succeed or fail in meeting donor and staff needs under real-world conditions such as after-hours use and semi-outdoor exposure. These comparisons inform design decisions by highlighting specific limitations in current solutions and guiding the development of a system that improves upon them.

a) Library Book Drop-off

Used to return books after hours through a sloped chute system. The narrow slot and angled entry prevent people from reaching in or climbing inside, improving security. However, the design does not cushion items as they fall, which limits protection for fragile materials.



Figure 1: Library Book Chute (image from [6])

b) Package Lock Boxes

Used to securely store delivered packages through a one-way locking mechanism. Once deposited, items cannot be removed without authorization. Some models include electronic locks and motion alarms that activate if tampering is detected. These systems are not optimized for large or delicate items or continuous deposits of items. They tend to get overflowed quickly.



Figure 2: Package Lock Box (image from [7])

c) Clothing Donation bins

Used for collecting clothing and shoes in large quantities. These bins provide strong weather resistance and high storage capacity, but their openings are often too high for some users to reach. The design also lacks impact protection, making it unsuitable for fragile or valuable donations, and theft remains a recurring issue, since people are still able to climb into the bin.



Figure 3: Donations Bin(image from [8])

The insights gained from this benchmarking analysis directly informed the subsequent concept generation process by defining which features needed to be retained, modified, or redesigned to better satisfy prioritized user needs

Table 3. Product Comparison

Criterion	Library Book Chute	Package Lockbox	Donations Bin
Security	X	✓	X
Fragile protection	X	X	X
Accessibility	✓	X	✓
Durability	✓	✓	✓

The comparison in Table 3 shows that none of the evaluated solutions adequately satisfy the most important user needs of security, protection of fragile items, accessibility, and durability. Because none of the existing solutions sufficiently meet all prioritized user needs, a new concept was required, directly motivating the concept generation and selection process described in the following section.

3.4 Concept Generation and Selection

To formally evaluate the leading concepts identified through sketching and prototyping, a weighted decision matrix was developed using criteria derived from the most critical user needs and engineering requirements. The evaluated concepts included gravity-based chute systems, single-door lockbox designs, and a two-door interlock system with controlled horizontal transfer. Criteria emphasized security, protection of fragile items, accessibility, durability, and system complexity. The decision matrix results are presented in Table 4.

3.4.1 Prototype 1: Early Feasibility Exploration

The first prototype was developed to assess basic mechanical feasibility and preliminary control logic. It began as a hand sketch and was constructed primarily from wood to enable rapid fabrication and iteration. This version used a roller-based exterior door and a scissor mechanism actuated by a single servo motor. Distance sensors were used to trigger door motion; however, sensing logic and control integration were limited, and the conveyor system was not yet incorporated

While Prototype 1 demonstrated the feasibility of a mechanically actuated deposit door, it exhibited inconsistent performance during repeated operation. The single-actuator scissor mechanism frequently jammed, and the roller system lacked precise motion control. These limitations highlighted the need for independent actuation, improved sensing logic, and more robust mechanical alignment. The shortcomings observed in Prototype 1 motivated the transition to alternative architectures in subsequent iterations.

3.4.2 Prototype 2: Two-Door Interlock and Controlled Transfer

Building on the lessons from Prototype 1, the second prototype introduced a two-door interlock architecture with each door driven by an independent stepper motor. This change enabled precise control of door sequencing and timing, directly addressing key accessibility and security requirements described in Section 3.2. The front face of the system was redesigned using 3D-printed plastic panels to improve water resistance, while the internal structure remained wood-based to support continued iteration.

Prototype 2 integrated a PIR sensor to detect human presence and an exterior button to initiate the donation sequence once the outer door closed. A level conveyor belt was incorporated for the first time, allowing donated items to be transferred horizontally into the collection area without vertical drops. The system followed a defined operational sequence, including object detection, human presence verification, door actuation, conveyor activation, and system reset. This prototype successfully demonstrated coordinated door control, sensing, and item handling, narrowing the viable design space to a small number of candidate concepts.

3.5 Concept Evaluation and Selection

To formally evaluate the leading concepts identified through prototyping, a weighted decision matrix was developed using criteria derived from the most critical user needs and engineering requirements. These criteria emphasized security, protection of fragile items through controlled transfer, accessibility, durability, and system complexity. The decision matrix is presented in Table 6.

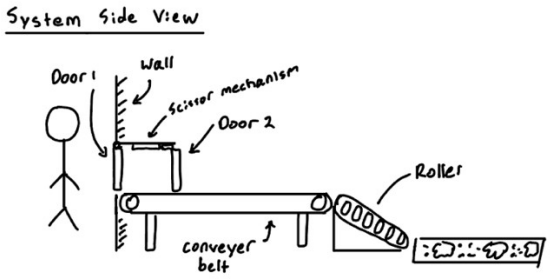


Figure 4: Prototype 1- Two-Door System (Side View Sketch)

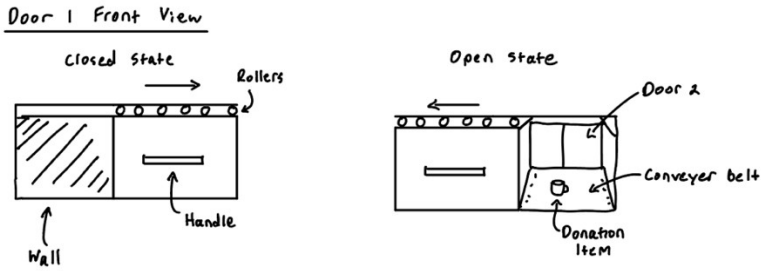


Figure 5: Prototype 1- Two-Door System (Outside Front View Sketch)

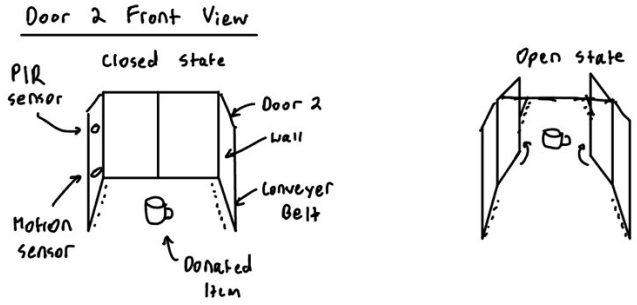


Figure 6: Prototype 1- Two-Door System (Inside Front View Sketch)

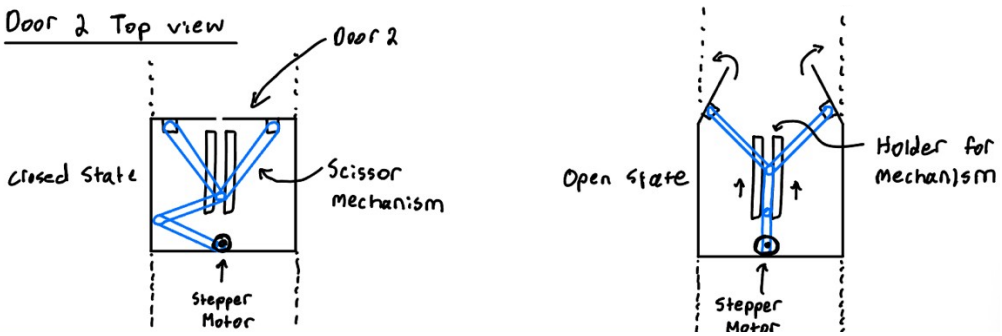


Figure 7: Prototype 1- Two-Door System (Interior Door Top View Sketch)

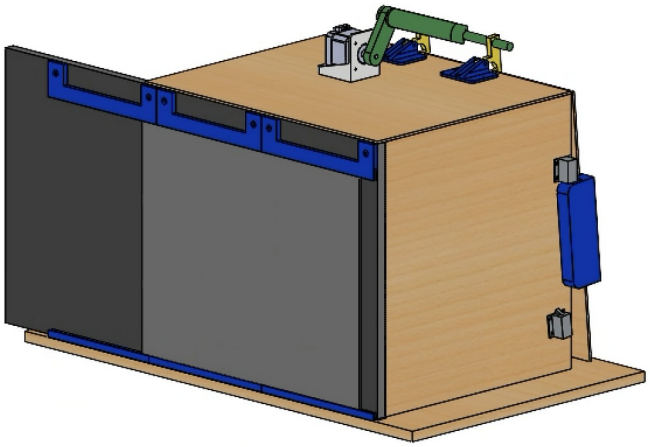


Figure 8: Early Prototype Assembly

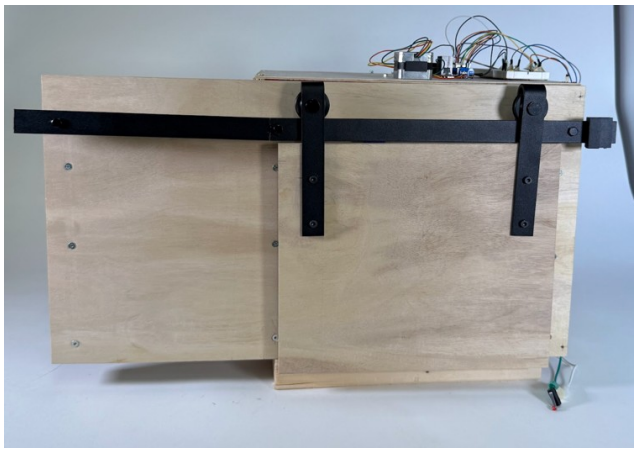


Figure 9: Prototype 1 Exterior View

Front view of the first physical prototype showing the donor-facing interface and enclosure layout.

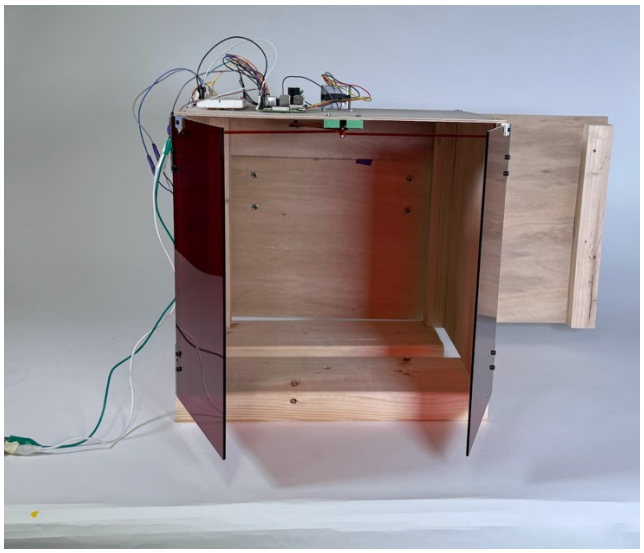


Figure 10: Prototype 1 Interior View

Rear view of the first physical prototype illustrating internal door alignment, conveyor placement, and wiring layout.

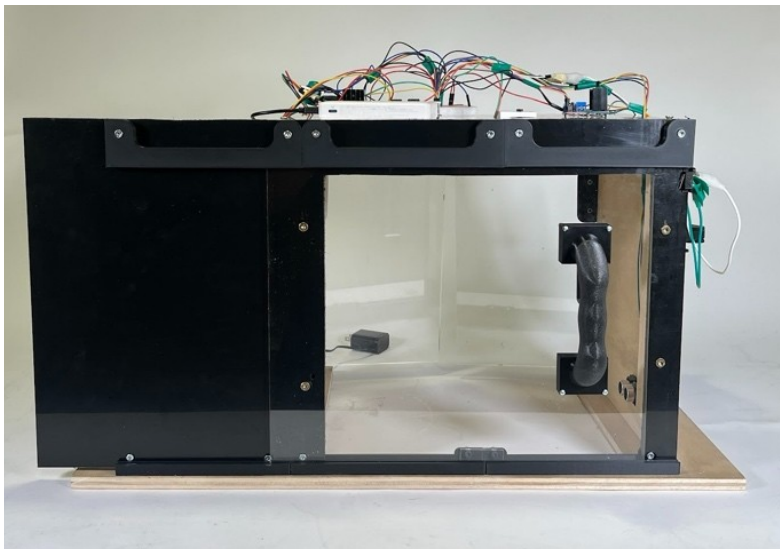


Figure 11: Prototype 2 Front View
Front view of the second prototype showing the fully enclosed, water-resistant donor-facing interface with integrated sensors.

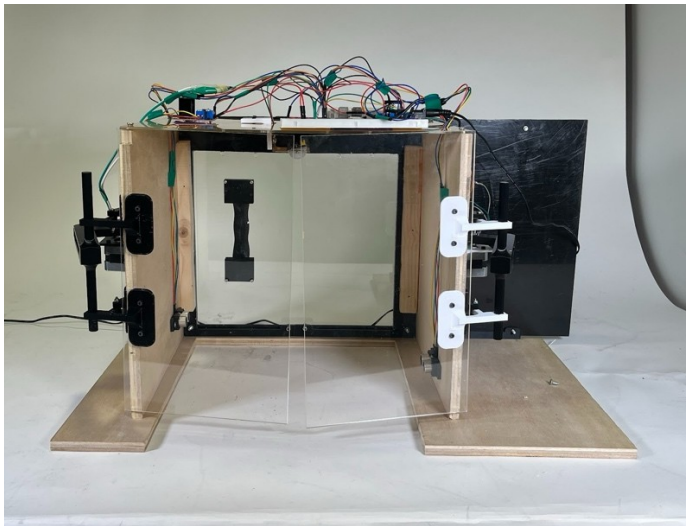


Figure 12: Prototype 2 Rear View
Rear view of the second prototype highlighting internal wiring, locking hardware, and electronics access.

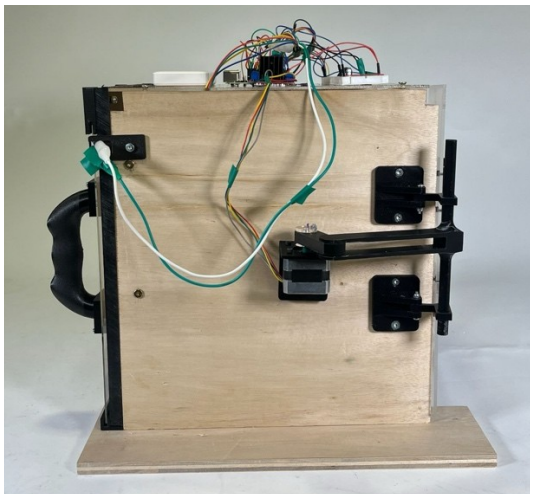


Figure 13: Prototype 2 Side View
Side view of the second prototype illustrating the two-door interlock layout and added sensor integration.

Table 4: Decision Matrix for Concept Selection

The concepts evaluated in the decision matrix include a gravity-based chute system, a single-door secure lockbox design, and a two-door interlock system with controlled horizontal item transfer via a level conveyor.

Scoring scale: 0 (poor) to 1 (excellent). Weighted score = Σ (criterion weight × score).

Criterion	Weight	Library Chute	Package Lockbox	Two-Door Interlock + Level Conveyor
Security (SEC)	0.30	0.60	0.90	0.90
Fragile Item Protection (FRG)	0.25	0.45	0.35	0.85
Accessibility (ACC)	0.20	0.90	0.70	0.80
Durability (DUR)	0.15	0.80	0.90	0.85
Complexity / Cost	0.10	0.90	0.75	0.65
Weighted Score	1.00	0.67	0.73	0.84

Based on the results of the decision matrix, the two-door interlock with a level conveyor transfer (Concept C) achieved the highest weighted score. This concept provided the strongest balance of controlled access, continuous item support, and reliable sequencing while remaining accessible and feasible to implement.

3.5.1 Iteration Toward the Final Integrated System

Following concept selection, a third prototype was developed to integrate enhanced security features, user feedback elements, and system monitoring into a complete, deployable system. This prototype focused on refining the selected architecture rather than exploring alternative concepts. The detailed design and

operation of the final integrated system are presented in the following sections. Following concept selection, targeted engineering analysis was conducted to validate key structural design decisions prior to finalizing the system architecture.

3.6 Engineering Calculations and Analysis

This section presents a finite element analysis (FEA) conducted to evaluate the effectiveness of the rear door locking configuration used in the after-hours donation system [1]. The primary objective of this analysis was not to predict failure loads, but rather to validate that the selected lock placement adequately constrains deformation and limits door opening under realistic forced-opening conditions. The system utilizes a double-door configuration with three lock contact points: two positioned at the lower left and lower right edges of the doors, and a third positioned at the top center along the seam between the two doors. This triangular lock arrangement was selected to reduce rotational motion and minimize separation at the central seam.

3.6.1 Modeling Approach and Assumptions

Due to the geometric complexity of the full latch hardware and enclosure assembly, a simplified finite element model was developed. This approach was selected after initial attempts to model the full system resulted in meshing instability caused by small features and complex contact interactions. The simplified model focuses on the rear door panels and lock contact locations, which are the dominant contributors to deformation behavior under loading.

The lock contact points were represented using fixed boundary conditions at the three physical lock locations. These fixed constraints represent engagement with a rigid enclosure frame and intentionally provide a conservative approximation of stiffness. Contact interactions were defined between the doors and each lock block to prevent penetration while allowing realistic load transfer. This modeling strategy allows for meaningful comparison of deformation modes while avoiding unnecessary numerical instability.

A static loading condition was applied to simulate a forced-opening scenario. A 50 lbf outward load was applied to each door in the direction of opening, representing a realistic manual push or pull applied during attempted misuse. This load was distributed across the door surfaces to avoid artificial stress concentrations from point loading.

3.6.2 Load Cases and Material Definitions

A static loading condition was applied to simulate a forced-opening scenario. A 50 lbf outward load was applied to each door in the direction of opening, representing a realistic manual push or pull applied during attempted misuse. This load was distributed across the door surfaces to avoid artificial stress concentrations from point loading.

To evaluate sensitivity to material stiffness and ensure that conclusions were not dependent on a single material assumption, two separate simulations were conducted. In the first case, the doors were modeled using cast acrylic (PMMA), matching the material used in the physical prototype. In the second case, the doors were modeled using steel to represent a stiff reference condition. All other boundary conditions and loading parameters were kept identical between the two studies to allow direct comparison.

3.6.3 Boundary Conditions and Simulation Setup

Our study illustrates the applied boundary conditions and loading configuration used in the analysis (see Figure 14). Fixed constraints were applied at the lower left and lower right lock locations, as well as at the top-center lock location positioned along the seam between the two doors. This configuration reflects the physical lock placement used in the final prototype.

The analysis was performed as a linear static study using SolidWorks Simulation. A curvature-based mesh was applied, with localized mesh refinement near lock contact regions to capture deformation behavior accurately. Although stress values near fixed constraints are sensitive to modeling assumptions, displacement and deformation patterns remain reliable indicators for evaluating lock placement effectiveness.

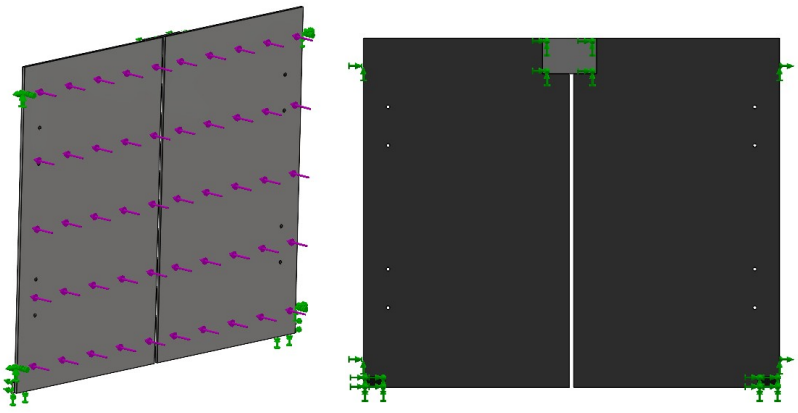


Figure 14: Finite Element Model Boundary Conditions
Simplified finite element model showing applied loads and fixed constraints at the three lock contact points.

3.6.4 Results and Interpretation

Figures 15 and 16 present the resultant displacement and von Mises stress distributions for the steel and cast acrylic door configurations, respectively. In both simulations, visible deformation occurs under the applied loading; however, no complete structural failure or uncontrolled door opening is observed.

This behavior indicates a tendency for the doors to rotate about the lower lock points, producing localized separation at the central seam. Importantly, the presence of the top-center lock significantly constrains this motion, preventing excessive opening and stabilizing the double-door system. Comparison between the steel and acrylic cases shows that while the magnitude of deformation increases for the acrylic material, the overall deformation pattern remains unchanged. This demonstrates that lock placement, rather than material stiffness alone, governs the system's structural response.

Although additional lock points or local reinforcements could further reduce seam deformation, the selected three-point triangular locking configuration provides an effective balance between security, simplicity, and manufacturability. The results confirm that the chosen lock positions are appropriate for the scope and functional requirements of the project.

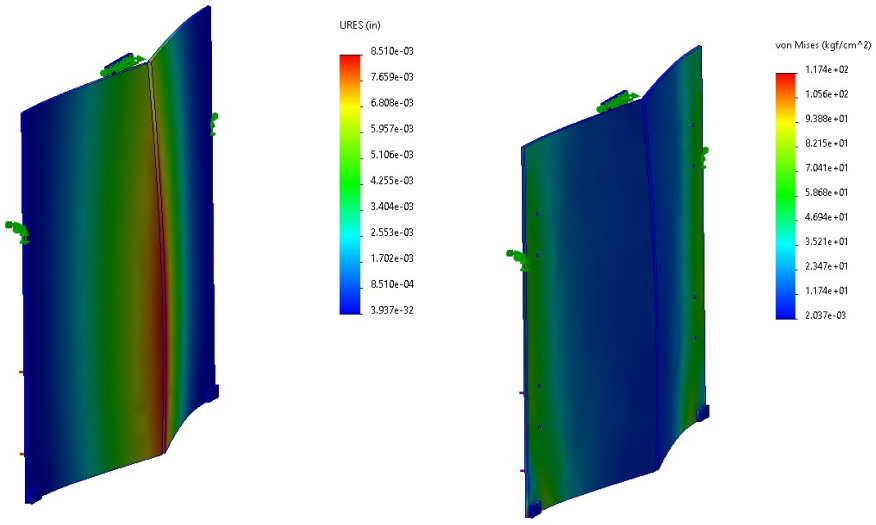


Figure 15: FEA Results for Steel Door Configuration
Resultant displacement and von Mises stress distributions for the interior doors modeled with steel material properties.

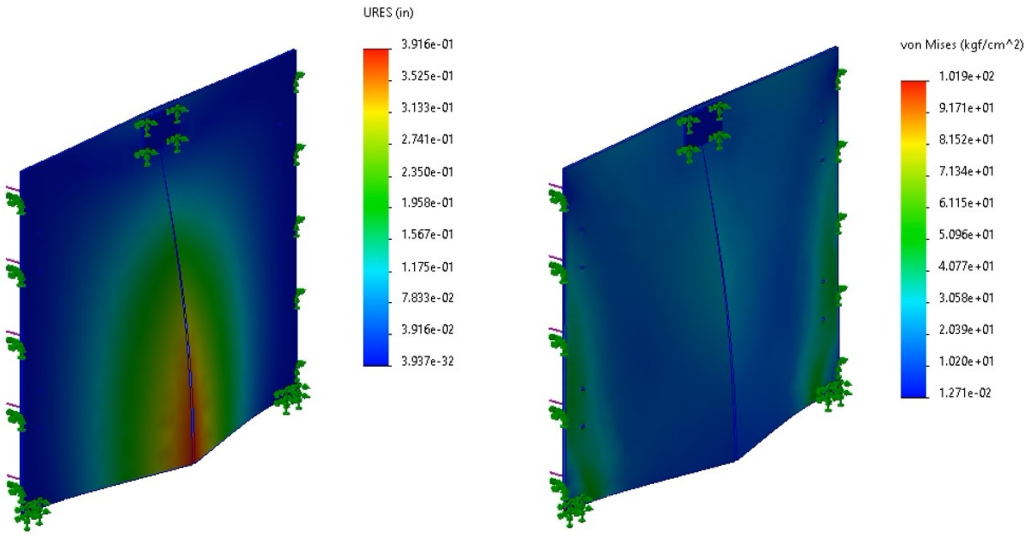


Figure 16: FEA Results for Acrylic Door Configuration
Resultant displacement and von Mises stress distributions for the interior doors modeled with cast acrylic (PMMA), matching the prototype.[1]

[1] This finite element analysis approach was developed based on methods learned in ME 41 (Engineering Design II) and reinforced through SolidWorks Simulation coursework and lab experience.

3.6.4 Design Implications

The findings from this analysis directly informed the final lock placement used in the prototype. By identifying the seam between the two doors as the primary region of deformation, the analysis validated the need for a top-center lock to supplement the two lower lock points. Future iterations of the design may incorporate localized stiffening near the seam or higher-fidelity latch modeling; however, the current configuration provides adequate structural constraint for the intended application.

3.7 Description of Final Design Solution

The final design is a secure and accessible after-hours donation system based on a two-door interlock architecture combined with a level conveyor transfer mechanism. The system enables one-way donation of items while preventing unauthorized access, protecting fragile goods through controlled motion, and providing clear feedback to users during after-hours operation. The design integrates mechanical, electrical, and sensing subsystems into a coordinated system that satisfies the user needs and engineering requirements defined in Sections 3.1 and 3.2. The overall configuration of the final prototype, including the enclosure, door layout, and support structure, is shown in Figures 17 and 18.



Figure 17: Final Prototype Front View
Front view of the final prototype showing the donor-facing exterior door, ergonomic handle placement, and sealed front panel designed for water-resistant operation.

3.7.1 System Architecture and Mechanical Layout

The system consists of an exterior donor-facing door, an interior security door, and an internal collection area connected by a level conveyor belt. The exterior door provides the user interface for depositing items and is positioned within ADA-compliant reach limits. Once closed, the exterior door cannot be reopened until the donation cycle is complete. The interior door, located behind the exterior door, remains locked until safety conditions are verified. The spatial relationship between the two doors and the enclosure is illustrated in the front and rear views of the final prototype shown in Figures 17 and 18.

Both doors are actuated independently using stepper motors, allowing precise control over door sequencing and timing. Independent actuation ensures that only one door is open at any time, preventing reach-in or retrieval of deposited items. The mechanical linkage used to actuate the interior security door is

shown in Figure 19. The enclosure is supported by a rigid frame that maintains alignment between moving components and provides structural stability. In a production configuration, structural components would transition from prototype materials to steel to improve durability and resistance to tampering while maintaining the same overall system architecture.

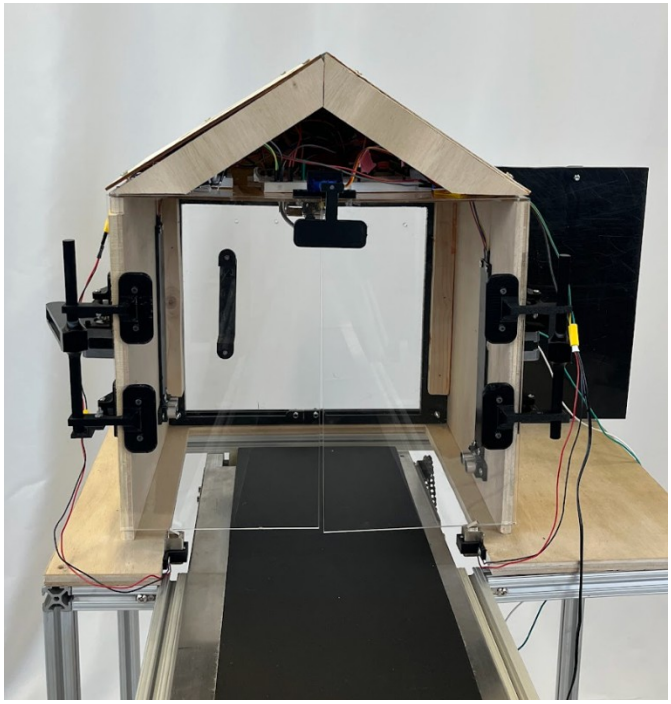


Figure 18: Final Prototype Rear View
Rear view of the final prototype highlighting locking hardware, electronics mounting, and wiring layout for system access and maintenance.

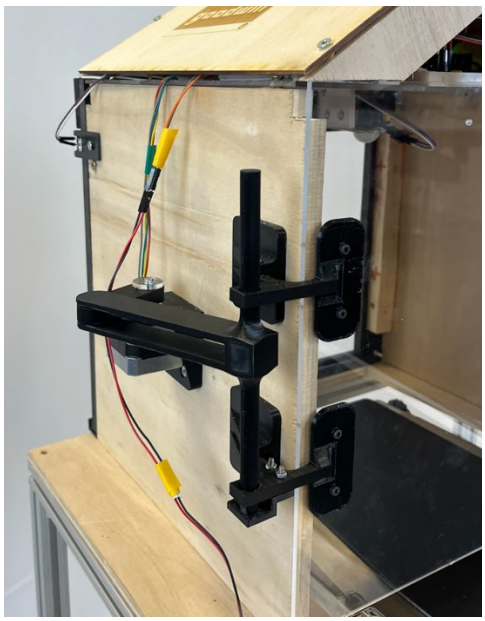


Figure 19: Door Actuation Mechanism Close-Up
Close-up view of the mechanical door arm and linkage used to actuate the interior security door during operation.

3.7.2 Item Transfer and Fragile Item Protection

Unlike traditional donation chutes that rely on vertical drops, the final design eliminates free fall entirely. Donated items are transferred horizontally using a level conveyor belt that maintains continuous support throughout the deposit process. This approach minimizes impact forces and significantly reduces the likelihood of damage to fragile items such as glassware and electronics. The conveyor layout and integration within the enclosure are shown in Figure 20.

The conveyor belt is activated only during an authorized donation cycle and transports items directly into the internal collection area. By controlling item motion rather than absorbing impact, the system provides predictable, low-energy transfer and reduces wear on internal components. This design choice also simplifies structural loading assumptions for downstream analysis and supports consistent performance during repeated after-hours use.



Figure 20: Level Conveyor Belt
Interior view showing the leveled conveyor belt used to transfer donated items horizontally without vertical free fall.

3.7.3 Sensing, Control Logic, and Safety Features

System safety is ensured through integrated sensing and control logic that monitors user presence and system state throughout operation. A PIR sensor detects human presence within the enclosure, preventing the interior door from opening if a person is detected. Limit switches provide position feedback for door state verification, and ultrasonic distance sensors are used to confirm item placement and clearance before actuation. The placement of these sensing components within the system is shown in Figure 21.



Figure 21: Sensor Integration Overview
Interior view highlighting the placement of the PIR sensor, limit switch, and ultrasonic distance sensors used for safety and control logic.

The system follows a defined operational sequence: detection of a deposited item, verification that no human presence is detected, unlocking and actuating the interior door, activating the conveyor belt, and resetting the system. Upon completion of each cycle, both doors return to their locked state and the system resets to a ready condition. This sequencing prevents unsafe operation and enforces one-way access throughout the donation process.

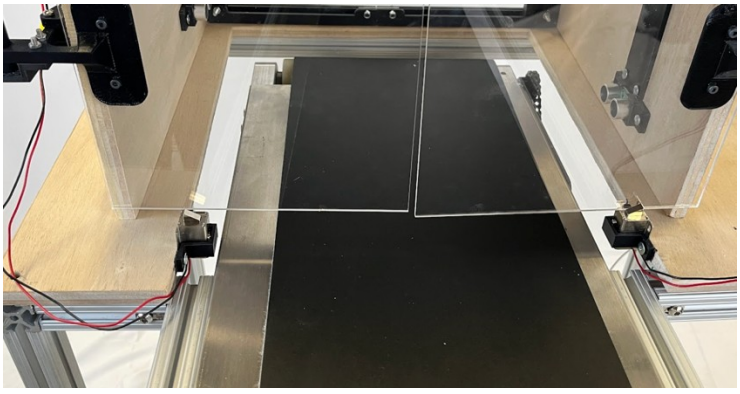


Figure 22: Bottom Lock Configuration
View of the bottom locking points used to resist prying and constrain deformation of the interior security door.



Figure 23: Top Servo Lock Mechanism
View of the top servo-actuated lock used to enforce the two-door interlock and prevent simultaneous access.

3.7.4 Ergonomic Handle Design and Accessibility

The exterior donor-facing door incorporates an ergonomic handle designed to support accessible, low-effort operation. The handle geometry provides a broad contact surface that accommodates different hand sizes and grip styles, reducing the need for fine motor control. The handle is positioned within ADA-compliant reach limits and is intended to be operable using a closed fist or palm. The handle design and placement are shown in Figure 24.

This design supports equitable access for a wide range of donors during after-hours operation, including users with limited grip strength or mobility. By minimizing required effort and simplifying interaction, the handle contributes to an intuitive and inclusive user experience.



Figure 24: Ergonomic Handle Design
Close-up view of the donor-facing handle designed for low-effort, accessible operation within ADA reach guidelines.

3.7.5 *User Feedback, Instructions, and Monitoring*

To support intuitive after-hours use, the system provides clear visual feedback and guidance to donors. A simple, three-step instruction set is displayed directly on the exterior of the system, guiding users through the donation process in a clear and accessible manner. This instructional signage is shown in Figure 25.

In addition to instructions, visible warning signage indicates that the system is protected by multiple sensors and automated mechanisms. This messaging discourages tampering while reassuring donors that their items are being handled securely. An external indicator light communicates system state, displaying a green signal when the system is ready to accept donations and a red signal when a donation cycle is in progress. Together, these feedback elements improve usability, reduce misuse, and increase donor confidence during after-hours operation.

An external indicator light communicates system state, displaying a green signal when the system is ready to accept donations and a red signal when a donation cycle is in progress. An internal white light illuminates the enclosure during operation, improving visibility in low-light conditions. For operational awareness, the system includes a donation counter that increments after each completed cycle and updates a web-based interface, allowing Goodwill staff to monitor system status and determine when emptying or inspection is required.



Figure 25: After-Hours Donation Instruction Poster
Exterior signage displaying step-by-step donation instructions and safety warnings for after-hours use.

3.7.6 Summary of Final Design

The final design integrates secure access control, controlled item transfer, and intuitive user feedback into a single after-hours donation system. The two-door interlock prevents unauthorized access, the level conveyor belt protects fragile items through continuous support, and the sensing and monitoring subsystems ensure safe and reliable operation. This architecture provides a scalable and deployable solution for Goodwill locations, with structural validation and performance testing presented in the following sections.

Although the final prototype was physically fabricated using prototype-grade materials, production-grade steel and aluminum were modeled in later CAD assemblies to evaluate enclosure geometry, structural layout, and long-term durability considerations for real-world deployment.

3.8 System Validation and Performance Testing

System validation was conducted to evaluate whether the final prototype meets key functional, safety, and usability requirements defined in Sections 3.1 and 3.2. Given the prototype-stage nature of the system and the scope of a senior capstone project, validation focused on functional behavior, safety logic, and user interaction rather than formal standards certification. Testing included repeated donation cycling, sensor lockout verification, and observational user interaction during a public demonstration.

3.8.1 Validation Overview and Requirement Mapping

Table 5 summarizes the primary validation activities conducted and the corresponding engineering requirements addressed. Where full-scale or standards-certified testing was not feasible, results are reported as scaled or observational validation, consistent with the prototype scope.

Table 5 Validation Tests and Results

Test	Requirement Validated	Target	Result
Continuous operation cycling	CAP+, SEC	≥ 24 hr operation without jamming (scaled test)	Pass (2 hr scaled test)
Door interlock sequencing	SEC	One door open at a time	Pass
Conveyor transfer test	FRG	Zero vertical free-fall	Pass
PIR safety lockout	SEC	Interior door disabled when presence detected	Pass
User feedback indicators	INT	Clear system state communication	Pass

3.8.2 Continuous Operation and Reliability Validation

To evaluate continuous operation and donation flow (CAP+), the system was run repeatedly for approximately one hour of continuous cycling during dedicated testing, followed by an additional two-hour public showcase period. During these periods, the system completed repeated donation cycles including exterior door operation, sensor verification, interior door actuation, conveyor transfer, and system reset.

Across both the controlled test and the extended showcase, no mechanical jamming, belt misalignment, or logic failures were observed. Multiple donated items were successfully transferred during the two-hour showcase without interruption, supporting the conclusion that the system can handle repeated after-hours use at the prototype scale.

While the engineering requirement specifies 24 hours of continuous operation, the successful one-hour continuous run combined with the two-hour showcase provides a reasonable scaled validation of system reliability, especially considering that the system was not operating at full storage capacity. Extended duration testing would be a logical next step for future iterations.

3.8.3 Safety, Interlock, and Control Logic Validation

Security and safety validation focused on confirming correct door sequencing and sensor-based lockouts (SEC). Throughout all observed operation, the two-door interlock consistently ensured that only one door was open at a time, preventing reach-in or item retrieval.

The PIR sensor was tested by intentionally entering the sensing zone during attempted operation. In all cases, the interior door remained disabled while presence was detected, validating the safety lockout behavior. This confirms that the system enforces one-way access and prevents unsafe operation during donation cycles.

The conveyor system was observed during repeated transfers and maintained continuous horizontal support of donated items, eliminating vertical free-fall and satisfying the fragile item protection requirement (FRG).

3.8.4 User Interaction and Instruction Validation

Usability and intuitive interaction (INT) were evaluated through observational user testing during system demonstration. Approximately 30 participants were asked to simulate a donation using only the posted three-step instructions, without verbal guidance.

At least 90% of users were able to complete the donation successfully without confusion or difficulty, indicating that the instruction set and visual feedback were effective. The indicator lights clearly communicated system state, reducing hesitation and discouraging interaction during active cycles.

One usability limitation identified was that the exterior handle visually suggested a pull motion, whereas the door operates via a sliding action. While this did not prevent successful use, it represents an opportunity for future improvement through revised labeling, handle geometry, or iconography to better match user expectations.

3.8.5 Validation Summary and Limitations

System validation demonstrated that the final prototype meets the primary functional objectives of secure after-hours donation, user accessibility, and protection of donated items. Core behaviors including two-door interlocking, sensor-based safety lockout, and controlled item transfer were validated through repeated operation and public interaction. During a two-hour public showcase, the system completed multiple donation cycles without jamming or unintended access, and all donations were successfully transferred from the exterior door to the secured interior space.

User interaction testing further supported the effectiveness of the design. Approximately 30 users simulated the donation process using posted instructions, and over 90 percent were able to complete a donation without assistance. This indicates that the exterior door (donor-facing) interface, visual feedback indicators, and instructional signage were sufficiently intuitive for first-time users.

Several limitations should be noted. Formal compliance testing for accessibility and environmental protection standards was not conducted. While the exterior door handle geometry was designed to align

with ADA ergonomic guidelines, door actuation force was not measured using a calibrated force gauge. Similarly, although enclosure features were designed with ingress protection in mind, formal IP54 dust and splash testing in accordance with IEC 60529 was not performed. Future validation would include quantitative measurement of door operating forces and controlled ingress testing to verify compliance.

Additionally, mechanical validation focused on functional performance rather than material certification. Components fabricated from prototype materials such as PLA and acrylic were sufficient for demonstrating system behavior, but long-term durability, wear, and impact resistance were not evaluated. These limitations do not detract from the validity of the system architecture but instead reflect the scope of a prototype-stage design.

Overall, the validation process confirms that the system architecture and control logic function as intended, while identifying clear pathways for future testing and refinement as the design transitions toward production-ready implementation.

4. Production Plan & Social/Ethical Considerations

4.1 Production Plan

For large-scale production, the prototype would transition from PLA and wood components to weather-resistant aluminum and steel to ensure long-term outdoor durability. Structural panels and frames would be fabricated using sheet metal forming and extrusion-based manufacturing, while brackets and mounts would be produced using CNC machining or laser cutting to maintain dimensional consistency. Standardized fasteners and modular panel interfaces would simplify assembly, installation, and maintenance across multiple Goodwill locations.

Electronics and sensing components would be housed in sealed enclosures rated for outdoor use and assembled as modular subassemblies to allow for straightforward replacement or upgrades. This manufacturing approach supports scalable production, cost control through batch fabrication, and adaptability to site-specific constraints, enabling efficient deployment across a range of Goodwill facilities.

4.2 Social & Ethical Considerations

This design directly affects community trust and equitable access to donation services. The system is designed to remain inclusive for users of different heights, strengths, and abilities by following ADA accessibility standards for reach and operating force [3]. Ethically, the design supports Goodwill's social mission by ensuring donations intended to help others are protected from theft and environmental damage, preserving resale value and reducing unnecessary waste [1].

Beyond donors and staff, nearby residents and passersby are indirect stakeholders. The installation must not obstruct sidewalks, generate excessive noise, or attract misuse. Low-noise operation and a compact footprint were considered to minimize disruption during after-hours use. Testing and iterative feedback from a diverse group of users help identify unintentional barriers or usability issues that may not be immediately apparent.

One limitation of the current prototype is that it has not yet been evaluated under all possible environmental and misuse scenarios. Extended outdoor testing and long-term monitoring would be necessary to identify edge cases such as extreme weather exposure or sustained vandalism. From an ethical design perspective, acknowledging these limitations ensures transparency and supports responsible deployment by highlighting areas for continued evaluation and improvement.

5. Conclusions and Future Work

5.1 Strengths of the Design Solution

The proposed after-hours donation system successfully demonstrates a functional and well-validated proof of concept that addresses the primary needs identified by stakeholders: secure after-hours donation, user accessibility, and protection of donated items. A major strength of the design is the two-door interlock architecture, which physically separates the exterior door (donor-facing) from the interior door (security door), preventing unauthorized access while maintaining a straightforward donation process.

The system effectively integrates mechanical design, sensing, and control logic to achieve reliable operation. Sensor-based lockout mechanisms ensure that only one door can be opened at a time, while the conveyor-based transfer mechanism eliminates vertical free fall, reducing the risk of damage to fragile items. Validation through continuous operation and a public showcase demonstrated robust performance, with no observed jamming and successful completion of multiple donation cycles.

From an engineering standpoint, targeted finite element analysis provided justification for key structural decisions, particularly the placement of lock contact points in the double-door configuration. Results showed that the selected three-point lock arrangement effectively constrains deformation and limits door opening at critical locations. Together, these features demonstrate that the system architecture is sound, scalable, and capable of meeting functional requirements with appropriate material selection.

5.2 Limitations of the Design Solution

While the prototype meets its primary functional goals, several limitations are acknowledged. First, formal standards-based testing was not conducted. Although the exterior door handle geometry was designed to align with ADA ergonomic guidelines, quantitative measurements of door actuation force were not performed using calibrated instrumentation. Similarly, while enclosure features were designed with environmental protection in mind, formal ingress protection testing in accordance with IP54 standards was outside the scope of the project.

Material selection also represents a limitation inherent to prototype-stage development. Several components were fabricated using PLA and acrylic, which are suitable for demonstrating system behavior but not intended for long-term outdoor use. Durability, wear, and impact resistance were not evaluated under extended environmental exposure or repeated loading cycles.

Additionally, the finite element analysis relied on simplified boundary conditions to evaluate deformation trends rather than predict exact failure loads. While this approach was appropriate for validating lock placement and system behavior, higher-fidelity modeling would be required for production-level certification.

5.3 Suggestions for Revisions to the Design Solution

Future revisions should focus on transitioning the system from a functional prototype to a deployable product. The most immediate improvement would be the substitution of prototype materials with production-grade alternatives, such as steel or aluminum for structural and locking components and impact-resistant polymers for door panels. These changes would significantly improve durability and environmental resistance without altering the overall system architecture.

Additional revisions should include formal validation testing. Door actuation forces should be measured and adjusted to ensure full compliance with ADA accessibility standards. Environmental testing, including dust and water exposure per IP54 guidelines, would further validate suitability for outdoor deployment. Extended lifecycle testing could be conducted to evaluate wear on moving components such as hinges, locks, and the conveyor system.

Design refinements may also include reinforcing high-stress regions identified through finite element analysis, improving sealing at door interfaces, and refining user-facing signage to further enhance clarity. With these revisions, the proposed system has strong potential to be implemented in real-world donation centers as a secure, accessible, and user-friendly after-hours solution.

AI Contribution

AI tools, specifically ChatGPT (OpenAI, 2025), were used solely for grammar correction, paragraph organization, and formatting assistance. All conceptual design ideas, engineering reasoning, and decision-making were developed entirely by the project team.

6. References

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7. Appendices- Technical Documentation

A. Matrix

A needs-to-requirements matrix was developed to verify traceability between user needs and engineering requirements. Each user need maps to at least one measurable requirement, ensuring complete coverage of stakeholder priorities.

Table 6: Matrix of user needs and engineering requirements

Need ↓ / Requirement →	Drop ≤ 6 in	Reach ≤ 48 in	Force ≤ 5 lbf	Anti-fish (Pass)	Tamper-resist (≥ 0.5 in + lock)	IP ≥ 54	Capacity 20 gal (5*)	Noise ≤ 55 dBA	CAP+ ≥ 24 hr
SEC — Security				X	X	X			
FRG — Fragile protection	X								
ACC — Accessibility		X	X						
DUR — Durability (outdoor)						X			
CAP — Capacity							X		X
CLN — Ease of Maintenance							X		X
NOI — Low Noise Operation								X	
UNB — Unobtrusive Design		X	X					X	
INT — Intuitive User Experience	X	X							
RPC — Standardizable		X	X				X		
CAP+ — Continuous Donation Flow							X		X

* Prototype scaled to 5 gallons due to build constraints.

B. Concepts

Below are sketches and concept illustrations explored during the ideation phase of the project. Each concept addresses key stakeholder needs, particularly security, protection of fragile items, and donor accessibility.

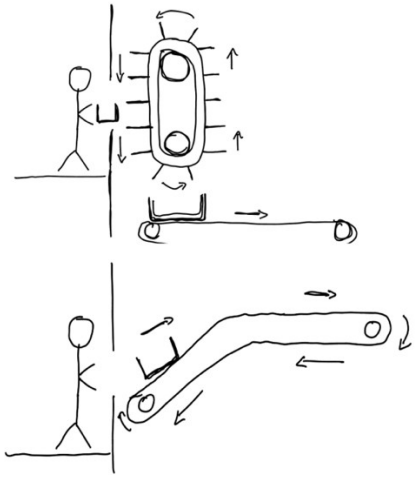


Figure 26: Vertical Conveyor Belt System Sketch

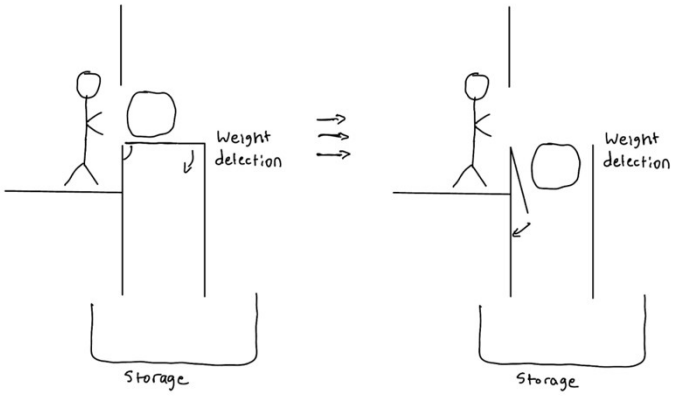


Figure 27: Classic Chute Drop System Sketch

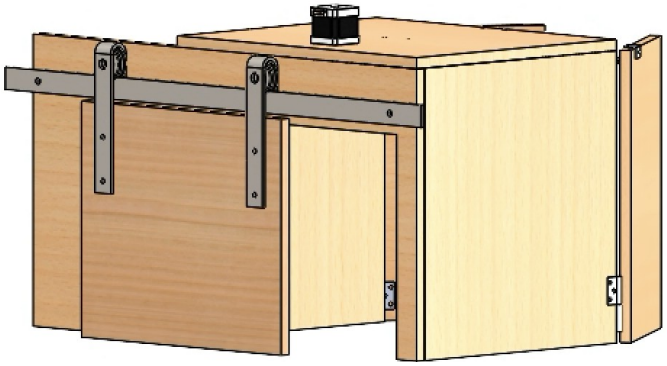


Figure 28: Prototype #1 Assembly

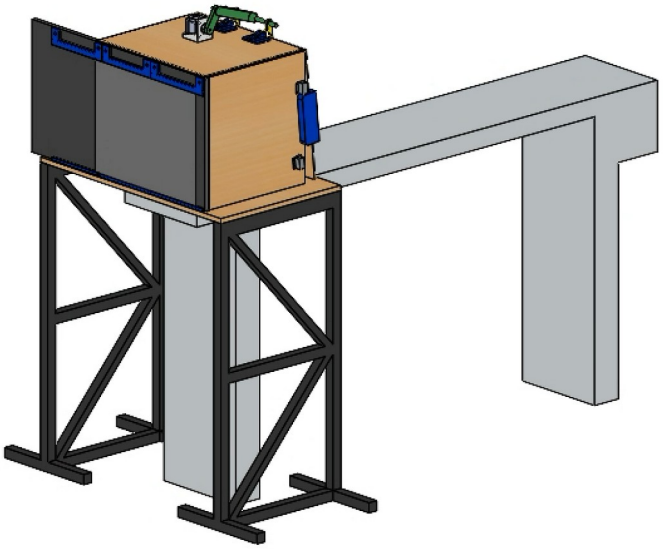


Figure 29: Prototype #2 Assembly

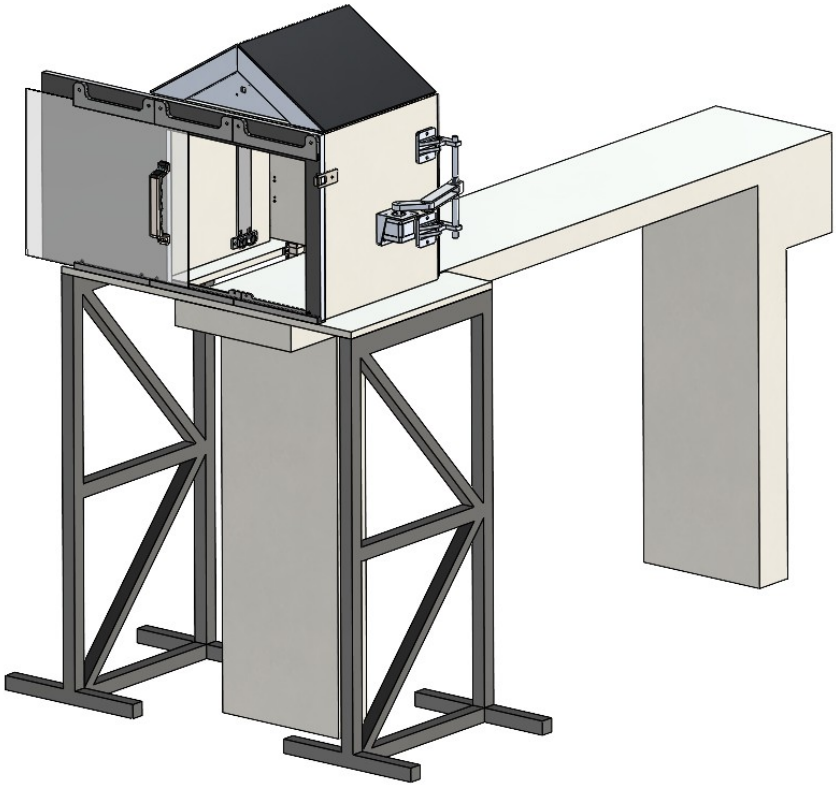


Figure 30: Final Prototype Full Assembly (Production-Grade Materials)

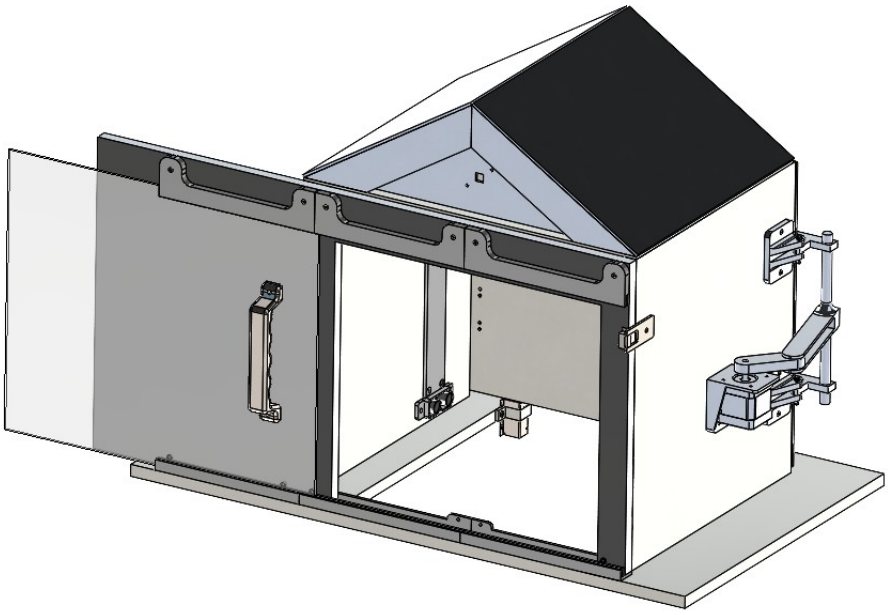


Figure 31: Final Prototype Housing Assembly

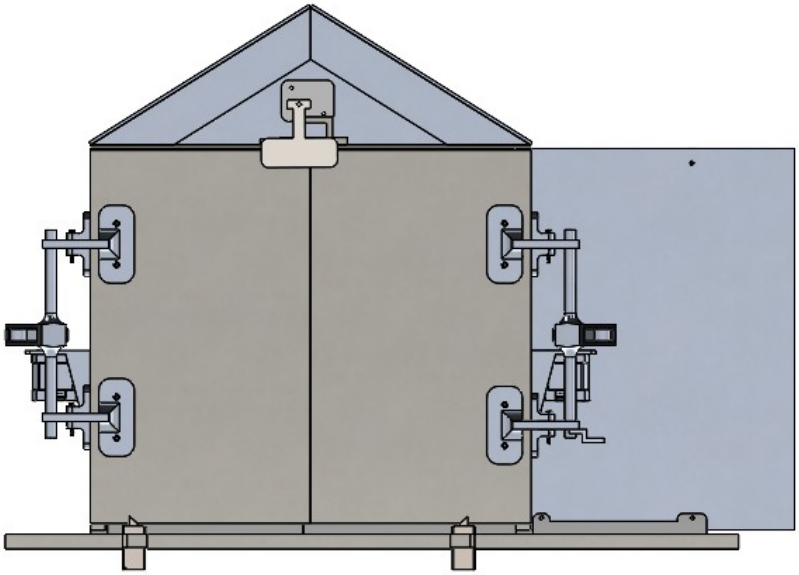


Figure 32: Final Prototype Housing Rear View

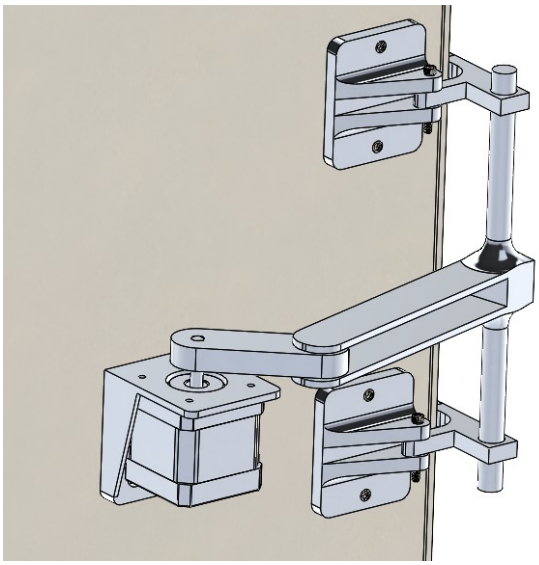


Figure 33: Door Actuation Mechanism (SolidWorks Model)

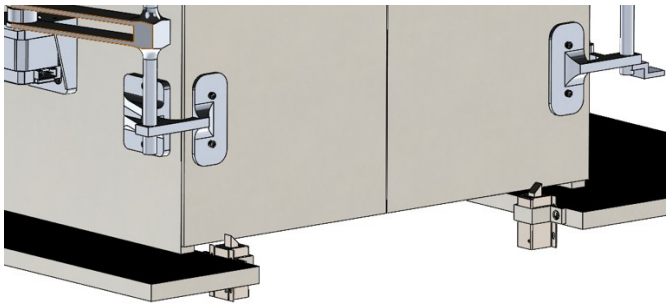


Figure 34: Bottom Locking Mechanism (SolidWorks Model)

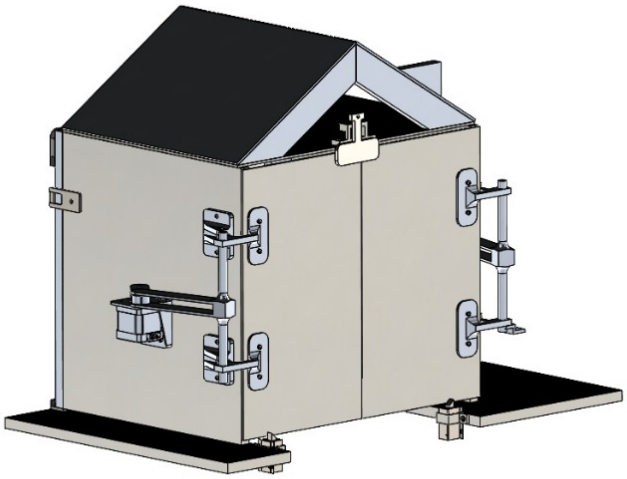


Figure 35: Final Prototype Additional Rear Perspective

C. Code Repository and Software Documentation

Control logic for the after-hours donation system was developed to manage sensor input, door actuation, locking behavior, and system state transitions during operation. To support transparency, reproducibility, and future development, all control code and supporting scripts have been documented and archived in a public GitHub repository.

GitHub Repository:
https://github.com/sbrizu/SeniorDesignProject

D. Engineering Drawings and Assembly Documentation

Engineering drawings were created to document the mechanical design of the after-hours donation system and to support fabrication, assembly, and future production considerations. These drawings capture both prototype-level geometry and production-intent configurations modeled using appropriate materials.

The drawing set includes detailed part drawings for major structural and mechanical components, as well as assembly drawings illustrating system layout, component interfaces, and relative positioning. Assembly drawings were developed to communicate the relationship between the exterior housing, doors, conveyor mechanism, locking hardware, and internal electronics. Together, these documents provide a clear reference for system construction, maintenance, and potential scale-up.

The following drawings were completed as part of this project:

1. Front Panel – Main Frame
2. Right Panel – Main Frame
3. Left Panel – Main Frame
4. Top Panel – Main Frame
5. Frame Hinge Connector
6. Door Arm
7. Door Hinge Connector
8. Motor Bracket
9. Slider Rail Bottom Left
10. Slider Rail Bottom Middle
11. Slider Rail Bottom Right
12. Slider Rail Top Left
13. Slider Rail Top Middle
14. Slider Rail Top Right
15. PIR Sensor Mount Holder
16. Roof Beam
17. Roof Inside Panel
18. Roof Top Cover
19. Bottom Lock Holder
20. Top Lock Holder
21. Door Arm Stopper
22. Wire Cover
23. Counter Switch Holder
24. Distance Sensor Holder
25. Light Holder Base
26. Light Holder Top
27. Interior Door
28. Front Door
29. Top Lock Arm
30. PIR Sensor Mount
31. Motor Arm Link
32. Side Frame
33. Right Frame
34. Long Frame
35. Left Frame
36. Height Frame
37. Length Frame

These drawings can be accessed on the following link.

The following assemblies and subassemblies were completed as part of this project:

1. Final Prototype Assembly
2. Main Frame Subassembly
3. Door System & Sliding Mechanism Subassembly
4. Actuation System Subassembly
5. Support Structure Subassembly

These drawings can be access on the following link.

E. System Demonstration Video

A system demonstration video documents both normal operation and safety behavior of the after-hours donation system. The video shows a successful donation cycle as well as the system response when human presence is detected by inserting an arm into the enclosure. When a human is detected, the status light turns red and begins flashing, all mechanical motion is inhibited, and an alert is sent to the system. No doors or actuators are engaged until the system is cleared and reset.

System Demonstration Video Link:
https://youtu.be/ZXyD4t2Kzso

F. Bill Of Materials

The bill of materials reflects the components used in the functional prototype of the after-hours donation system. Listed items include structural components, actuation and locking hardware, sensing elements, and electronics required to demonstrate full system operation. Materials and components were selected to support rapid prototyping, iterative testing, and validation of system behavior rather than long-term outdoor deployment. As a result, certain production-grade components and environmental sealing elements are not fully represented. A production-ready bill of materials would incorporate alternative materials and additional components appropriate for long-term durability, standards compliance, and large-scale deployment.

Bill of Materials						
Product:				Date: November 11, 2025		
Assembly: Main Frame and Enclosure						
Item #	Part Number	Qty	Description	Material	Vendor	Est. Cost (USD)
1.	FP-001	1	Front Panel – Main Frame	PVC Sheet	Amazon	40.99
2.	RP-001	1	Right Panel – Main Frame	Plywood	Home Depot	91.10
3.	LP-001	1	Left Panel – Main Frame	Plywood	Home Depot	15.00
4.	TP-001	1	Top Panel – Main Frame	Plywood	Home Depot	15.00
5.	RB-001	2	Roof Beam	Wood	Home Depot	10.00
6.	RIP-001	1	Roof Inside Panel	Plywood	Home Depot	12.00
7.	RTC-001	2	Roof Top Cover	Plywood	Home Depot	12.00
Assembly: Door System and Sliding Mechanism						
8.	ACR-CLR-062-12-24	1	Front Door	Acrylic (PMMA)	Amazon	3.00
9.	ACR-CLR-062-12-24	2	Interior Door	Acrylic (PMMA)	Amazon	6.00
10.	DH-001	1	Door Handle	PLA	3D Printed	n/a

11.	FHC-001	4	Frame Hinge Connector	PLA	3D Printed	n/a
12.	DA-001	2	Door Arm	PLA	3D Printed	n/a
13.	DHC-001	4	Door Hinge Connector	PLA	3D Printed	n/a
14.	DAS-001	1	Door Arm Stopper	PLA	3D Printed	n/a
15.	SR-TL-001	1	Slider Rail – Top Left	PLA	3D Printed	n/a
16.	SR-TM-001	1	Slider Rail – Top Middle	PLA	3D Printed	n/a
17.	SR-TR-001	1	Slider Rail – Top Right	PLA	3D Printed	n/a
18.	SR-BL-001	1	Slider Rail – Bottom Left	PLA	3D Printed	n/a
19.	SR-BM-001	1	Slider Rail – Bottom Middle	PLA	3D Printed	n/a
20.	SR-BR-001	1	Slider Rail – Bottom Right	PLA	3D Printed	n/a
21.	MAL-001	1	Motor Arm Link	PLA	3D Printed	n/a
Assembly: Conveyor System						
22.	B0C2C8ZJ6Y	1	Conveyor Belt	Rubber	Amazon (Vevor)	9.99
23.	CBB-001	1	Conveyor Belt Base	Plywood	Home Depot	10.00
Assembly: Actuation and Locking Hardware						
24.	SY42STH47-1206A	2	Stepper Motor (NEMA 17)	Steel	Amazon	25.98
25.	MB-001	2	Motor Bracket	PLA	3D Printed	n/a
26.	TLH-001	1	Top Lock Holder	PLA	3D Printed	n/a
27.	ARM-001	1	Top Lock Arm	PLA	3D Printed	n/a
28.	SOLH-001	2	Bottom Lock Mount	PLA	3D Printed	n/a
Assembly: Sensors, Electronics, and UI						
29.	B07KBWVJMP	1	PIR Sensor	—	Amazon	8.69
30.	PIR-MH-001	1	PIR Sensor Mount Holder	PLA	3D Printed	n/a
31.	DSH-001	2	Distance Sensor Holder	PLA	3D Printed	n/a
32.	LSH-001	1	Limit Switch Holder	PLA	3D Printed	n/a
33.	LS-001	1	Limit Switch			
34.	WC-001	2	Wires Cover	PLA	3D Printed	n/a
35.	LHB-001	2	Light Holder Base	PLA	3D Printed	n/a
36.	LHT-001	1	Light Holder Top	PLA	3D Printed	n/a
37.	B01MZ7J2NQ	2	Ultrasonic Distance Sensor (HC-SR04)	—	Adafruit	30.00
38.	SRV-001	1	Servo Motor	—	Amazon	8.00
39.	SRV-ARM-001	1	Servo Motor Standard Arm	—	Amazon	n/a
40.	SL-001	2	DC Solenoid Lock, 12V	—	Amazon	24.00
41.	MC-001	3	Motor Controller, L298N Dual H-Bridge	—	Amazon	15.00
42.	RM-001	1	Relay Module, 5V Single Channel	—	Amazon	5.00
43.	LED-001	2	NeoPixel RGBW LED (SK6812)	—	Amazon	12.00
44.	VC-001	1	DC-DC Voltage Converter (Buck)	—	Amazon	8.00
45.	LLS-001	1	Logic Level Shifter (5V–3.3V)	—	Amazon	4.00
46.	RES-330	2	Resistor, 330 Ω, 1/4 W	—	Amazon	0.50
47.	Aypzuke	1	Breadboard	—	Amazon	9.99

	B0BTT48V7P					
48.	SIM&NAT B07RX78T9L-F2F	10	Jumper Wires Female–Female	—	Amazon	3.00
49.	SIM&NAT B07RX78T9L-M2M	10	Jumper Wires Male–Male	—	Amazon	3.00
50.	SIM&NAT B07RX78T9L-M2F	10	Jumper Wires Male–Female	—	Amazon	3.00
51.	RPI-4B	1	Raspberry Pi (Model 4B or equivalent)	—	Amazon	55.00
Assembly: Support Structure and Fasteners						
52.	40-8020-3M	2	80/20 Aluminum Extrusion	Aluminum	McMaster-Carr	57.86
53.	91251A110	16	Black-Oxide Alloy Steel Socket Head Screw	Steel	McMaster-Carr	4.00
54.	91251A106	12	Black-Oxide Alloy Steel Socket Head Screw	Steel	McMaster-Carr	5.00
55.	90044A111	4	Black-Oxide Alloy Steel Socket Head Screw (as listed)	Steel	McMaster-Carr	1.50
56.	91251A113	10	Black-Oxide Alloy Steel Socket Head Screw	Steel	McMaster-Carr	1.50
	Team Member: Ariana Olivares			Prepared By: Ariana Olivares		
	Team Member: Nathan Henderson			Checked By: Nathan Henderson		
	Team Member: Sol Brizuela			Approved By: Sol Brizuela		
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